

Trusted Aunti

A safety-aware assistant for adolescent girls in Kenya: contraception, sexual health, staying safe, and getting to a service.

Build report — the refined version.

github.com/George-techie/girl-effect-srh-rag · 166 tests · 8 governed sources, 1,693 chunks · 8 verified services across 8 routes

What the brief asked for

The previous submission was assessed as *overengineered in places*, with a request to see what a more refined version focused on delivery of the use case would look like. The scope also narrowed: **sexual and reproductive health, centred on contraception and family planning**, plus the empowerment outcomes that follow. Mental health and menstruation left the topic list.

This build is the answer to both. Same solution philosophy. Every component either justified by a measurement or deleted. **Refined does not mean smaller** — two components were added during the work, each because something measurable failed without them.

\$0.0109 COST PER TURN	52/52 ROUTING ACCURACY	1.000 SAFEGUARDING RECALL & PRECISION	24/24 GROUNDING ANSWERS CITING
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Performance against the previous build

All previous figures below are read from that project's own recorded evaluation runs of 4 August 2026, 51 cases each, not reconstructed.

CONFIGURATION	ACC.	UNSAFE	UNHELPFUL	CALLS/TURN	TOKENS/TURN	COST/TURN	MEDIAN LATENCY
A — no safeguarding layer	0.784	7	1	0.98	3,938	—	5,476 ms
B — safeguarding, no evidence judge	0.882	0	5	2.65	3,848	—	7,058 ms
C — B plus the LLM evidence judge	0.745	0	12	3.53	6,838	—	7,030 ms
B+ — full profile, five model roles	0.843	1	6	3.78	8,599	\$0.0189	12,410 ms
This build	52/52†	—	3	0.69	4,946	\$0.0109	3,536 ms

† Not the same measure. The previous action accuracy was scored by an LLM judge over 51 cases against a different scope and corpus; this build's 52/52 is deterministic routing accuracy. The columns that are directly comparable are calls, tokens, cost and latency — all measured over 51 and 52 cases respectively, priced with the same OpenRouter table.

On the refusal column: the previous build's "unhelpful actions" were scored by an LLM judge; this build's 3 are counted deterministically — blocked by the validator, no evidence found, or a provider error. Not the same instrument, so read it as the same order of magnitude rather than a precise ranking. The eight out-of-scope declines are excluded, because declining a question the service deliberately does not cover is correct behaviour rather than a refusal.

5.5× FEWER MODEL CALLS PER TURN	1.7× FEWER TOKENS PER TURN	42% LOWER COST PER TURN	38% OF TURNS NEED NO MODEL
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What that comparison does and does not support

Cost and consumption are the robust numbers. They are properties of the architecture and do not move with network conditions. **0.69 model calls per turn against 3.78** — a 5.5× reduction — and **4,946 tokens per turn against 8,599**, which prices at **\$0.0109 per turn against \$0.0189**. Twenty of the fifty-two messages reached no model at all, and those are the turns where it matters most: every safeguarding reply is assembled from approved text and verified table rows in **0 ms**.

Per-turn call rate varies with the mix of questions. On the 52-message decision set, which is deliberately heavy in safeguarding and out-of-scope cases, it is 0.69. On the 38-turn conversation set, which is mostly questions, it is 1.03. Both are reported rather than the flattering one.

Latency is a weaker claim and is presented as one. The previous project's own two runs of the identical System B recorded 3,993 ms and 7,058 ms — a 3,065 ms spread from provider variance alone. Against B+ the improvement is large and consistent with removing most of the model calls; against B it is inside the noise. The defensible statement is that this build does the same work in roughly one call instead of four, and latency followed.

The two accuracy columns are not directly comparable: the previous action accuracy was scored by an LLM judge over 51 cases against a different scope and a different corpus. This build's equivalent is 52/52 on deterministic routing, measured differently and reported separately below.

The ablation that shaped this build

The clearest finding in the previous project's own data is what the evidence judge cost. Comparing B with C, which differ only by that component:

- Action accuracy **0.882** → **0.745**, a drop of 13.7 points
- Unhelpful refusals **5** → **12**, more than doubled
- Unsafe actions **0** → **0**. It prevented nothing in that run

A component that refuses seven more girls and prevents no additional harm is not a safety feature. That single comparison is the strongest argument in this project for deletion over addition, and it came from the previous build's own measurements rather than from taste.

Architecture

validate → safeguarding screen → route → resolve* → prepare* → retrieve → generate → check

** conditional. Neither is a universal stage.*

Only generation uses a hosted LLM. Routing, safeguarding, resolution, query preparation and validation are deterministic rules. Embeddings run locally on `bge-m3` , so no question leaves the machine to be encoded and there is no per-query embedding cost.

The safeguarding screen is a gate, not a route. It runs first on every message, and when it fires nothing downstream executes: no retrieval, no model call, no validator. That ordering is the single most important decision in the build, because retrieval cannot decline — a deliberately out-of-scope question was measured retrieving at 0.668, above most in-scope ones.

Two output contracts, chosen by provenance

	GROUNDED	CONVERSATIONAL
Used for	factual, access	greetings, support, her ambitions
Safe because	every claim carries a citation	it makes no claim at all
Blocked on	an uncited claim	a citation marker with no passage

Which contract applies is decided by which path produced the turn, never inferred from whether citations happen to be present.

Detect broadly, escalate narrowly

One safeguarding route, two severities, implemented as one boolean. **Urgent** — force, threat, assault, contraceptive sabotage, self-harm — puts verified contacts in front of her. A **concern** — pressure, conditional consent, *"he keeps asking"* — is recognised as safeguarding and answered as a conversation, with contacts offered rather than pushed. Treating every coercion signal as an emergency both reads as being passed on when she came to talk, and at scale buries the services in cases that were never emergencies. Nothing contacts anyone on her behalf.

Evaluation

Five sets, 159 labelled items, each with criteria fixed before the run.

1 • Decision layer — 52 messages

METRIC	RESULT	NOTE
Overall accuracy	52/52	1.000
Safeguarding recall	12/12	1.000
Safeguarding precision	12/12	1.000
Contrast pairs	6/6	same surface form, different path

2 • Retrieval — 31 questions, tagged by Theory-of-Change driver

METRIC	NATURAL	PREPARED	ORACLE
Adequate@5 — a passage that <i>answers</i> her	0.880	0.960	1.000
Hit@5	0.926	0.963	0.926
MRR	0.883	0.920	0.864
Agency drivers, mean MRR	0.611	0.750	0.889

Gold labels are source-level, so Hit@5 is an optimistic upper bound and is reported as one. Questions are tagged by the eight behavioural drivers rather than generic RAG categories, because knowledge is one driver of eight: this retriever scored knowledge MRR 1.000 while self-identity sat at 0.417.

3 • Multi-turn — 4 journeys, 23 turns

Forbidden passages 1 → 0. Before conversation state existed, "*and does it hurt?*" asked after a question about the implant retrieved **female sterilization**, and "*where can I go?*" after a coercion disclosure retrieved **BTL**, which is permanent. Path accuracy 21/23; the two disagreements are disputed labels, ours, left uncorrected and documented.

4 • Mixed-intention messages — 15 messages

Wanted evidence 8 → 9 of 14, unwanted passages 4 → 3, and a purely emotional control that now retrieves nothing at all. Clause-level retrieval, with no word list deciding which clause is the health question — the similarity scores do that.

5 • Conversation quality — 19 journeys, 38 turns

PROPERTY	RESULT
Grounded turns that cite a source	24/24 • 100%
Reply matches her register	38/38 • 100%
Usable reply, not a refusal	38/38 • 100%
Free of a deferral	38/38 • 100%
Free of machinery talk	37/38 • 97%
Free of a passive standing offer	36/37 • 97%
Gives her somewhere to go next	32/37 • 86%

There is no LLM judge, and warmth is not measured. Every property above is defined in advance and checkable by reading the text, so the score means the same thing on every run and any row can be verified by hand. Warmth cannot be measured honestly, and the previous build used a judge for exactly that — it refused a girl's compliment, twice.

Getting her to a service

The Theory of Change runs behavioural drivers → intent → **service access** → behaviour change. A system that answers every question well and never gets her to a service has done the easy half.

HER TURN	WHAT SHE GETS
"Where can I get family planning near me?"	the cited answer, plus Marie Stopes and One2One with numbers
"Where can I go for an HIV test?"	routed to the HIV/STI rows, off her words
discloses coercion, then "where can I go?"	the help pathway, o model calls
self-harm risk	crisis lines in front of her, not behind a tap
pressure without force	contacts offered behind the tap, not pushed

Contacts are read from a verified table and **never generated**. No corpus chunk contains a phone number, so a number in generated text came from the model's memory, and the validator treats that as fatal. A row reaches a girl only once it carries a source, a checker and a date.

What the work actually found

Every defect below was found during this build. Several were invisible to the evaluation sets and surfaced only when a person typed a real sentence into the demo — which is the strongest argument in this report for user testing.

DEFECT	WHY IT MATTERED
Self-harm detected by a flag nothing set	The pipeline tested a tag against a list of regex source strings. Every self-harm disclosure phrased in English received the ordinary safeguarding reply instead of the urgent one with crisis numbers. Nothing errored.
The phone check fired on page numbers	Fatal, so an otherwise correct cited answer mentioning "p. 116" would have been refused. Found by verifying a claim written in the README.
A 48-word message classified as small talk	It opened with "thanks". Her HIV question never reached the corpus.
Support turns held to the grounded contract	The corpus has no answer to "I'm scared". It returned a passage about rape at 0.627 to a girl who said she was nervous about the clinic.
A word list fitted to one failing example	Tested on five phrasings written afterwards it fired on two, and one miss retrieved a passage about rape. Replaced with clause-level retrieval.

Known limitations

- **All five evaluation sets were written in-house.** They are a regression harness and a design instrument, not evidence about real girls.
- **78% of the corpus is provider guidance** — 1,326 clinical chunks against 58 written for a young reader. The facts are right; the intended reader is a clinician.

- **Asking in Kiswahili costs 0.062 similarity**, measured over five matched pairs. Direct translation is handled; idiomatic Sheng is harder.
- **Continuity sits at 86%**, the weakest conversation metric and the most recently added.
- **Two multi-turn labels are disputed**, ours, and left standing rather than corrected after seeing the result.

What was deliberately not built

No LLM evidence judge, no output judge, no turn planner, no model-based query rewriter as a first resort, no conversation summariser, no entity tracker, no per-girl profile, no external monitoring vendor. Observability is a JSONL file holding operational fields only — no message text, no identifier for her — because an event log for a safeguarding product, kept by default, is a surveillance database with a dashboard on it.

Trusted Aunti · build report · figures from the project's own evaluation runs. Previous-build numbers:

`evaluation/results/three_system_v2.json` and `system_bplus.json`, 4 August 2026, 51 cases each.